

🔍 Level 1: Python Basics — 1–20

1. Print "Hello, World!"
 2. Take two integers and print their sum.
 3. Take two integers and print sum, difference, product, and quotient.
 4. Find the square and cube of a number.
 5. Convert Celsius to Fahrenheit.
 6. Convert Fahrenheit to Celsius.
 7. Calculate the area of a circle.
 8. Calculate the area and perimeter of a rectangle.
 9. Calculate the average of three numbers.
 10. Swap two numbers.
 11. Check whether a number is positive, negative, or zero.
 12. Check whether a number is even or odd.
 13. Find the maximum of two numbers.
 14. Find the maximum of three numbers.
 15. Check whether a year is a leap year.
 16. Check whether a character is a vowel or consonant.
 17. Check whether a person is eligible to vote.
 18. Calculate simple interest.
 19. Calculate compound interest.
 20. Convert seconds into hours, minutes, and seconds.
-

🔍 Level 2: Loops & Number Problems — 21–40

21. Print numbers from 1 to N.
22. Print numbers from N to 1.
23. Print all even numbers from 1 to N.
24. Print all odd numbers from 1 to N.
25. Find the sum of numbers from 1 to N.
26. Find the sum of all even numbers from 1 to N.
27. Find the sum of all odd numbers from 1 to N.
28. Find the factorial of N.
29. Print the multiplication table of a number.
30. Count the number of digits in an integer.
31. Find the sum of digits of a number.
32. Reverse an integer.
33. Check whether a number is a palindrome.
34. Check whether a number is prime.
35. Print all prime numbers between 1 and N.
36. Count the number of prime numbers between 1 and N.
37. Find the GCD of two numbers.
38. Find the LCM of two numbers.

39. Check whether a number is an Armstrong number.
 40. Print all Armstrong numbers from 1 to N.
-

□ Level 3: Patterns — 41–50

41. Print a square pattern of *.
42. Print a right triangle pattern.
43. Print an inverted right triangle.
44. Print a pyramid pattern.
45. Print an inverted pyramid.
46. Print a number triangle.
47. Print Floyd's Triangle.
48. Print Pascal's Triangle.
49. Print a diamond pattern.
50. Print a hollow square pattern.

Example for #42:

```
*
**
***
****
*****
```

□ Level 4: Strings — 51–65

51. Count the number of characters in a string.
 52. Count vowels and consonants.
 53. Count uppercase and lowercase letters.
 54. Reverse a string.
 55. Check whether a string is a palindrome.
 56. Count the frequency of each character.
 57. Find the first non-repeating character.
 58. Find the first repeating character.
 59. Remove all spaces from a string.
 60. Remove duplicate characters from a string.
 61. Check whether two strings are anagrams.
 62. Find the longest word in a sentence.
 63. Find the shortest word in a sentence.
 64. Count the number of words in a sentence.
 65. Find the most frequent character in a string.
-

□ Level 5: Lists / Arrays — 66–80

- 66. Find the maximum element of a list.
 - 67. Find the minimum element of a list.
 - 68. Find the sum of all elements.
 - 69. Find the average of list elements.
 - 70. Count even and odd numbers.
 - 71. Reverse a list without using `reverse()`.
 - 72. Find the second-largest element.
 - 73. Find the second-smallest element.
 - 74. Remove duplicates from a list.
 - 75. Count the frequency of each element.
 - 76. Find the missing number from $1 \dots N$.
 - 77. Find duplicate elements in a list.
 - 78. Rotate an array left by K positions.
 - 79. Rotate an array right by K positions.
 - 80. Merge two sorted arrays.
-

□ Level 6: Searching & Sorting — 81–90

- 81. Implement Linear Search.
 - 82. Implement Binary Search.
 - 83. Implement Bubble Sort.
 - 84. Implement Selection Sort.
 - 85. Implement Insertion Sort.
 - 86. Implement Merge Sort.
 - 87. Implement Quick Sort.
 - 88. Find the K-th smallest element.
 - 89. Find the K-th largest element.
 - 90. Sort an array containing only 0, 1, and 2.
-

● Level 7: Intermediate / Competitive Programming — 91–100

- 91. **Two Sum**
Given an array and target κ , find two elements whose sum is κ .
- 92. **Maximum Subarray Sum**
Find the maximum sum of a contiguous subarray.
- 93. **Prefix Sum Queries**
Answer multiple range-sum queries efficiently.

94. **Longest Increasing Subsequence**

Find the length of the longest strictly increasing subsequence.

95. **Coin Change**

Find the minimum number of coins needed to make a given amount.

96. **0/1 Knapsack**

Maximize the value while keeping total weight within capacity.

97. **Fibonacci using Dynamic Programming**

Solve Fibonacci using memoization and tabulation.

98. **Graph BFS**

Given a graph, print its BFS traversal.

99. **Graph DFS**

Given a graph, print its DFS traversal.

100. **Dijkstra's Shortest Path**

Given a weighted graph and source vertex, find the shortest distance to every vertex.